Department of Electrical and Electronics engineering
Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. EEE
Course Code & Title: EE8702 Power System Operation and Control
Name of the Faculty member (s): Dr.S.Kannan

Innovative Practice Description

- **Unit / Topic:** Unit I / Preliminaries on PSOC
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 1
- **Activity Chosen:** Concept Map
- **Justification:**

Setting goals, solving problems, and creating action plans may all be done creatively with concept mapping. It allows you to swiftly jot down ideas in a free-form format. When a group uses mini-mapping, each participant's thoughts quickly stimulate ideas in others. Breaking free of old patterns to discover fresh and inventive methods is encouraged by this dynamic group interaction. For a better understanding of this topic, multiple types of magnetic materials and their features can be mapped onto a single map.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

After teaching the concept, the students were made to pair with their neighbors

Reporter: Myself

At the end the Class (Last 6 minutes)

1. I asked the students to think about Preliminaries on PSOC concept for 2 minute.
2. Then I told them to Pair with their neighbors and discuss about the concepts for another 1 minute.
3. Finally I told them to design mini-map and submit within 3 minutes.

- **CO – PO / PSO mapping:**

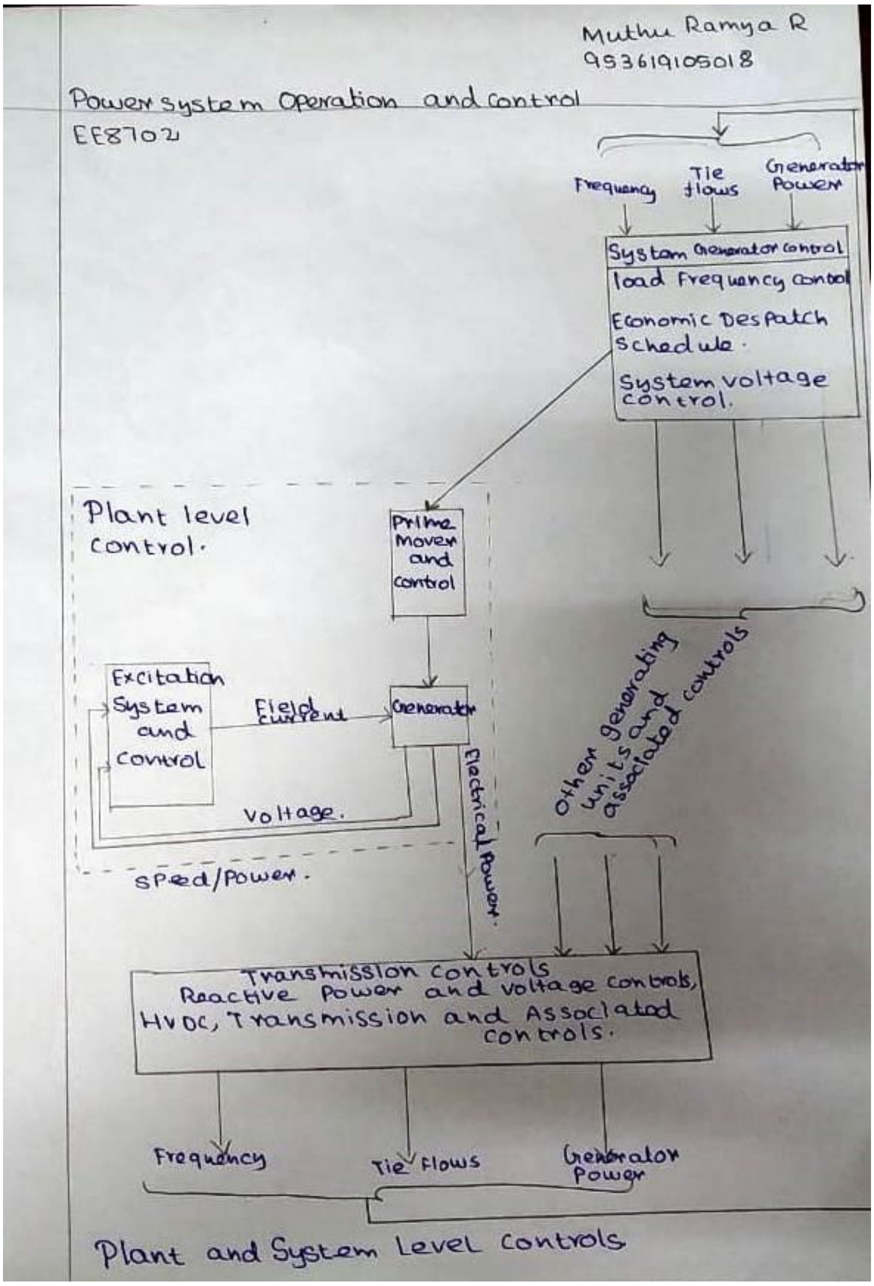
CO	PO1	PO2	PO3	PO4	PO8	PO10	PO12	PSO1	PSO2	PSO3
C402.1	2	3	1	1	1	2	1	2	3	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO8	PO10	PO12
	2	**1**	**2**
Justification for correlation	Student follow moral and ethical principles during their Exam and while completing the assignment	Student read, understand and interpret the given problem statement and able to produce clear and well-constructed documents.	Students able to engage in independent life-long learning in the broadest context of technological change in the control strategies in power system

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

    Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

  - ***Benefit of the practice:***

    Mini Maps are an excellent tool for students to take notes on what they learn. It allows students to jot down only the most significant details using key words, and then visually connect facts and concepts — all on one sheet. It made it easier for pupils to take crucial notes because it condensed pages of notes into a single sheet of paper. Slow learners were also able to recall facts more rapidly because to the small map.

Challenges faced in implementation:

In most cases, teachers will provide more detailed explanations in the topic's notes section. The students are divided into groups and asked to construct a map to show how the issues in the biomass conversion process are related. The activity was only supposed to last 10 minutes. However, in the actual world, this action takes 15 minutes to perform.

References:

1. https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-concept-maps

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Innovative Practice Description

- **Unit / Topic:** Unit II / Load Frequency Control
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 7
- **Activity Chosen:** Laboratory Demonstration
- **Justification:**

The Load Frequency control is one of the important operations in power system, when the load on the generator increases, which causes a decrease in frequency, this concept is demonstrated in alternator which is available in Electrical Machines II laboratory. After teaching the concept, I thought of conducting this activity for making the students to give practical exposure, which enhance the learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

After teaching the concept, I asked the students to give one or two minutes to think about the topic without writing anything.

Total Strength is 29

Photographer: one student – Mr.S.Jeyasurya (interested in photography)

Reporter: Myself

At the end the Class (Last 15 minutes)

- ✓ I asked the students to think about the load frequency control in grid operation for 2 minutes.
- ✓ Then I told them to Pair with their neighbours and discuss about the load frequency control for another 1 minute.
- ✓ Finally, I explained the importance of LFC in real world.

- **CO – PO / PSO mapping:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO10	PO12	PSO1	PSO2	PSO3
	2	2	2	1	1	2	1	2	2	1
Justification for correlation	Student explain the concept of tie line with frequency bias control by using fundamental engineering concepts.	Student identify the mathematical, engineering and other relevant knowledge that applies to analyze the static and dynamic analysis of single area system for controlled and uncontrolled cases	Student recognize the need for the analysis of static and dynamic analysis of two area system for controlled and uncontrolled cases. Students integrate the economic dispatch control with LFC for economic operation of power system	Student able to integrate the economic dispatch control with LFC for economic operation of power system by examining relevant method and analysis	Student follow moral and ethical principles during their Exam and while completing the assignment.	Student read, understand and interpret the given problem statement and able to produce clear and well-constructed documents.	Students able to engage in independent life-long learning in the broadest context of technological change in the control strategies in power system	Student solve the numerical calculation associated in constructing power-frequency controller to single area and two area systems using MATLAB.	Student able to design, prototype the power-frequency controller to single area and two area systems.	Student able to design and develop the power-frequency controller hardware required for single area and two area systems.

- **Images / Screenshot of the practice:**


- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

    Students told that it is good see all the operations of the alternator connected to the grid and this demonstration helps the students to understand the concepts easily.

- ***Benefit of the practice:***

1. Students can able to understand the impact of engineering solution on society
2. Most of the students attended the question asked Internal Assessment Test – I retest.
3. The success of the activity was evaluated by asking the same question during load frequency control (Unit – II) **– Around 80% of students answered.**
4. Students can able to explain the concepts in examination without any confusion.

- ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

- Nagrath I.J. and Kothari D.P., 'Modern Power System Analysis', Tata McGraw-Hill, Fourth Edition, 2011.
- Kundur P., 'Power System Stability and Control, Tata McGraw Hill Education Pvt. Ltd., New Delhi, 10th reprint, 2010.

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Innovative Practice Description

- **Unit / Topic:** Unit III / Reactive Power Voltage control
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 13
- **Activity Chosen:** Laboratory Demonstration
- **Justification:**

  The Reactive Power Voltage control is one of the important operations in power system, when the excitation of the alternator changes, which causes a change in voltage, this concept is demonstrated in alternator which is available in Electrical Machines II laboratory. After teaching the concept, I thought of conducting this activity for making the students to give some practical exposure, which enhance the learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

  (A brief content about how the practice was followed)

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO10	PO12	PSO1	PSO2	PSO3
C402.3	3	2	2	2	1	2	1	2	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO10	PO12	PSO1	PSO2	PSO3
	3	2	2	2	1	2	1	2	2	1
Justification for correlation	Student explain the concepts of reactive power	student identify the mathematical, engineering and other	Student recognize the need	Student apply appropriate instrumentation to make measurements	Student follow moral and ethical principles	Student read, understand and interpret	Students able to engage in independent life-long learning in	Student analyze various control actions to	Students design the suitable control	Students able to design and develop the hardware

	control for a power system and explain the operation of methods of voltage control using fundamental engineering knowledge.	relevant knowledge that applies to improve the stability of AVR loop for using stability compensation	to analyze the operation of various types of excitation system to form an Automatic Voltage regulator (AVR) also to analyze the static and dynamic analysis of it	of physical quantities using electrical machines, passive and active compensating devices	during their Exam and while completing the assignment.	the given problem statement and able to produce clear and well-constructed documents.	the broadest context of technological change in the control strategies in power system	maintain the voltage profile against various loads using modern software packages.	actions to maintain the voltage profile against various loads	and software required for maintaining the voltage profile against various loads.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***

Students told that it is good see all the operations of the alternator connected to the supply and this demonstration helps the students to understand the concepts easily.

- ***Benefit of the practice:***

1. Students can able to understand the impact of engineering solution on society
2. Most of the students attended the question asked Internal Assessment Test – I retest.
3. The success of the activity was evaluated by asking the same question during load frequency control (Unit – II) **– Around 80% of students answered.**
4. Students can able to explain the concepts in examination without any confusion.

- ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

- Nagrath I.J. and Kothari D.P., ‘Modern Power System Analysis’, Tata McGraw-Hill, Fourth Edition, 2011.
- Kundur P., ‘Power System Stability and Control, Tata McGraw Hill Education Pvt. Ltd., New Delhi, 10th reprint, 2010.

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Innovative Practice Description

- **Unit / Topic:** Unit IV / Unit Commitment
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 18
- **Activity Chosen:** Matlab Simulation
- **Justification:**

Unit commitment is the process of deciding when and which generating units at each power station to start-up and shut-down. By learning this student will able to decide what the individual power outputs should be of the scheduled generating units at each time-point to minimize the fuel cost.

- **Time Allotted for the Activity:** 20 minutes
- **Details of the Implementation:**
  - Unit Commitment problems were explained and the students were taken to PSS Lab.
  - A sample simulation were simulated in the lab and their doubts were clarified
  - A Problem was given to the students and asked to simulate individually and their simulation outputs were individually verified
  - Students submitted their work as a report and it is evaluated.

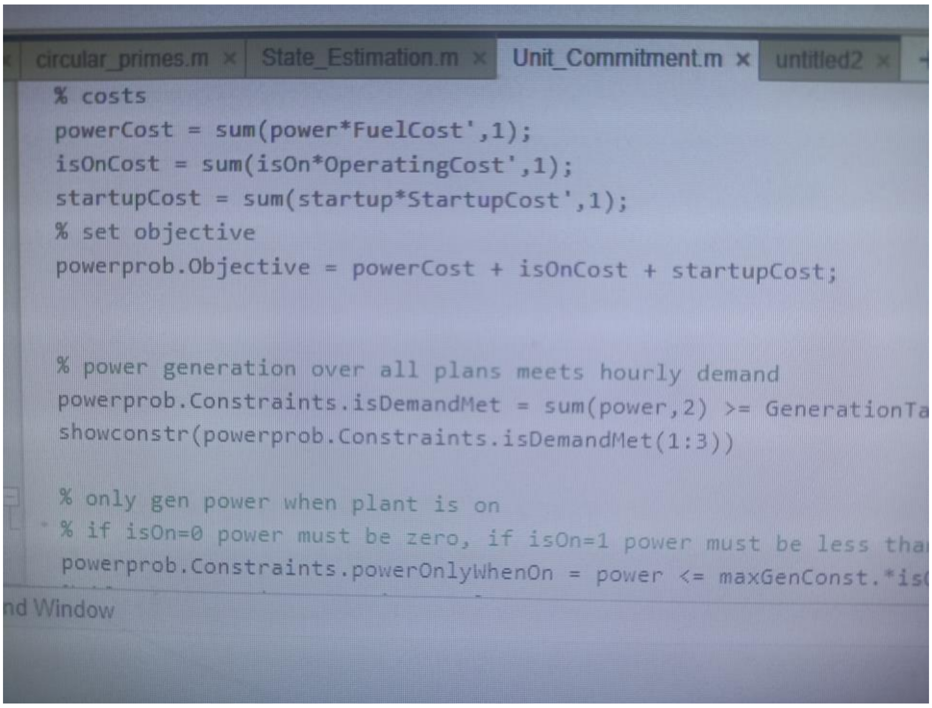
- **CO – PO / PSO mapping:**

CO	PO2	PSO1
C402.4	**2**	2

- **PO / PSO mapped:**

Innovative practice	PO2	PSO1
	2	**2**
Justification for correlation	Student identify the mathematical, engineering and other relevant knowledge to solve the unit commitment problem with constraints using priority list method	Student able to analyze the scheduling of the power system economically using modern software packages.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - While solving Student felt that their computational time was reduced due to simulation and they were able to solve many problems quickly with the help of simulation.
  - ❖ ***Benefit of the practice:***
    - Student understood the unit commitment problem clearly and able to explain the procedure of calculation whenever asked.
    - Most of the students have attended the question in their Internal Assessment Test.
    - Student Interest towards Matlab Simulation is increased
  - ❖ ***Challenges faced in implementation:***
    - Initially Few students took more time for their simulation and later by practice their time for completing the simulation is reduced

References:

https://in.mathworks.com/matlabcentral/fileexchange/32073-unit-commitment-by-dynamic-programming-method

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Innovative Practice Description

- **Unit / Topic:** Unit V / State Estimation
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 25
- **Activity Chosen:** Matlab Simulation
- **Justification:**
  State estimation is used in all Energy Management Systems (EMS) to identify the present operating state of a system. The bus voltage magnitude, real power injections, reactive power injections, active power flow, reactive power flow and line current flows are common measurements available in SCADA systems. By learning this student will analyze the underlying hardware and software structure which supports power system operation
- **Time Allotted for the Activity:** 20 minutes
- **Details of the Implementation:**
  - State Estimation problems were explained and the students were taken to PSS Lab.
  - A sample simulation were simulated in the lab and their doubts were clarified
  - A Problem was given to the students and asked to simulate individually and their simulation outputs were individually verified
  - Students submitted their work as a report and it is evaluated.

- **CO – PO / PSO mapping:**

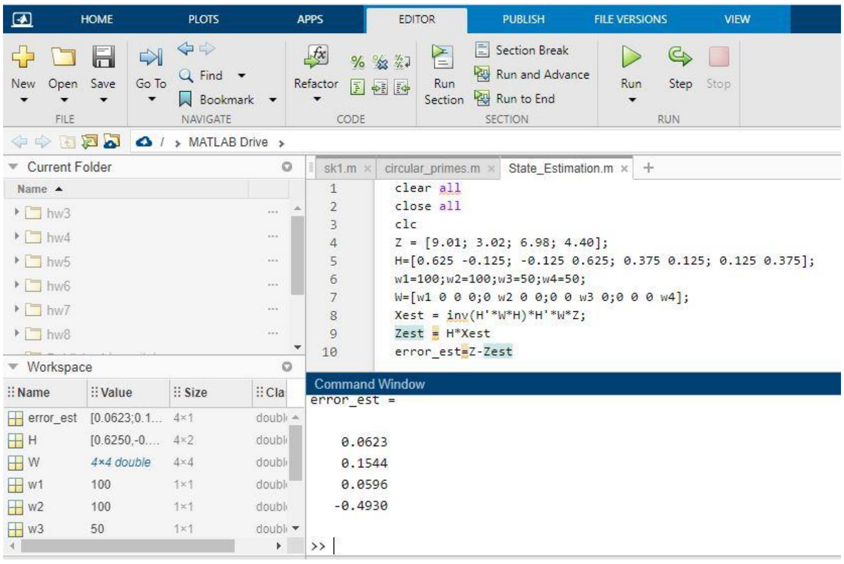
CO	PO3	PSO1
C402.5	2	2

- **PO / PSO mapped:**

Innovative practice	PO3	PSO1
	2	1
Justification for correlation	Student able to analyze the state estimation problem with measurements and errors and generate information through	Student analyzes the performance of computer control of power system using modern software packages.

	appropriate tests.	

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - While solving Student felt that their computational time was reduced due to simulation and they were able to solve many problems quickly with the help of simulation.
  - ❖ ***Benefit of the practice:***
    - Student understood the State Estimation problem clearly and able to explain the procedure of calculation whenever asked.
    - Most of the students have attended the question in their Internal Assessment Test.
    - Student Interest towards Matlab Simulation is increased
  - ❖ ***Challenges faced in implementation:***
    - Initially Few students took more time for their simulation and later by practice their time for completing the simulation is reduced

References:

1. https://www.researchgate.net/profile/Mohamed-Mourad-Lafifi/post/Does-anyone-have-code-preferably-for-Matlab-for-a-Generalized-State-Estimation/attachment/59d64a7a79197b80779a4c2c/AS%3A475444337221632%401490366171153/download/Power+Systems+State+Estimation_Final.pdf